1 Data Generation

1.1 Pattern Matching

A *pattern* is a set p. If a set, vector, element (or other data) d maches the pattern p, then $\Omega \in p$. The set $\{p \,|\, \Omega \in p\}$ is the set of patterns to which some data Ω belongs.

1.1.1 Example

$$\begin{array}{lcl}
\mathfrak{E} & := & \mathfrak{T} \cup \{\langle \omega_1, \cdots, \omega_n, \text{“+”}, \omega_i, \cdots, \omega_j \rangle \,|\, \langle \omega_1, \cdots, \omega_n \rangle, \langle \omega_i, \cdots, \omega_j \rangle \in \mathfrak{E}\} \\
\mathfrak{T} & := & \mathfrak{F} \cup \{\langle \omega_1, \cdots, \omega_n, \text{“}\times\text{”}, \omega_i, \cdots, \omega_j \rangle \,|\, \langle \omega_1, \cdots, \omega_n \rangle, \langle \omega_i, \cdots, \omega_j \rangle \in \mathfrak{T}\} \\
\mathfrak{F} & := & \mathfrak{N} \cup \{\langle \text{“(”}, \omega_1, \cdots, \omega_n, \text{“)”} \rangle \,|\, \langle \omega_1, \cdots, \omega_n \rangle \in \mathfrak{E}\} \\
\mathfrak{N} & := & N \cup N^* \\
N & := & \{\text{“0”}, \cdots, \text{“9”}\}
\end{array}$$

$$\begin{array}{lcl}
s & = & \langle \text{“(”}, \text{“5”}, \text{“+”}, \text{“1”}, \text{“0”}, \text{“)”}, \text{“}\times\text{”}, \text{“3”}, \text{“+”}, \text{“9”} \rangle \\
 & = & \langle \text{“(”}, \mathfrak{n}_1, \text{“+”}, \mathfrak{n}_2, \mathfrak{n}_3, \text{“)”}, \text{“}\times\text{”}, \mathfrak{n}_4, \text{“+”}, \mathfrak{n}_5 \rangle \\
 & = & \langle \text{“(”}, \mathfrak{n}_1, \text{“+”}, \mathfrak{n}_6, \text{“)”}, \text{“}\times\text{”}, \mathfrak{n}_4, \text{“+”}, \mathfrak{n}_5 \rangle \\
 & = & \langle \text{“(”}, \mathfrak{e}_6, \text{“)”}, \text{“}\times\text{”}, \mathfrak{n}_4, \text{“+”}, \mathfrak{n}_5 \rangle \\
 & = & \langle \mathfrak{f}_1, \text{“}\times\text{”}, \mathfrak{n}_4, \text{“+”}, \mathfrak{n}_5 \rangle \\
 & = & \langle \mathfrak{f}_1, \text{“}\times\text{”}, \mathfrak{f}_2, \text{“+”}, \mathfrak{n}_5 \rangle \\
 & = & \langle \mathfrak{t}_1, \text{“}\times\text{”}, \mathfrak{t}_2, \text{“+”}, \mathfrak{n}_5 \rangle \\
 & = & \langle \mathfrak{t}_3, \text{“+”}, \mathfrak{n}_5 \rangle \\
 & = & \langle \mathfrak{e}_3, \text{“+”}, \mathfrak{e}_5 \rangle \\
 & = \mathfrak{e} &
\end{array}$$

Pattern matching is the same as tokenwize parsing?